

Road traffic noise impact assessment model and its application based on route Noise Impact index

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ABSTRACT

With the continuous advancement of urbanization in China and the rapid expansion of road transportation networks, the contradiction between road operation noise and sensitive areas such as hospitals and schools has become more and more prominent. Traditional noise assessment methods mainly rely on the equivalent continuous A sound level (Leq) index, which is difficult to fully reflect the noise tolerance characteristics, affected building area, and affected population of different building types in the road noise-affected area, resulting in a large deviation between the noise impact assessment in the early planning stage and the actual operating noise. In order to solve these problems, a comprehensive evaluation model based on route noise impact index (RNII) is constructed, which incorporates the building acoustic environment sensitivity classification, affected building area, affected population and other indicators along the road into the comprehensive evaluation system, and constructs a noise impact index calculation formula with spatial distribution characteristics, which can perform refined noise impact analysis on sensitive targets along the planned route. Taking a highway project in Guangxi as an empirical case, the practicability of the model is verified by comparing the two comparison route schemes in the north-south direction. This road traffic noise impact assessment method based on route noise impact index can provide a scientific basis for the optimization of line position in the early stage of road engineering, and has positive practical significance for improving the traffic noise environmental impact assessment technology and promoting the acoustic environment protection in territorial spatial planning.

Keywords: Traffic noise; Environmental impact assessment; Route Noise Impact Index (RNII); Sensitive areas; Spatial distribution

1. INTRODUCTION

With the implementation of China's "National Transportation Power" policy and the rapid construction of road infrastructure, new societal concerns have emerged. As a result of transportation expansion, road noise has gradually become the second most major danger after air pollution, posing serious health risks to the general public^[1-3]. Prolonged exposure to noisy environments can cause sleeplessness, hypertension, and cardiovascular problems. Construction noise has a temporary influence on residents in noise-sensitive locations, but operating noise is a long-term worry. Conflicts between operational road noise and noise-sensitive protection targets such as residential areas and schools are becoming more common, particularly in highway projects, where operational noise impact assessment has evolved into a critical indicator for assessing the rationality of highway alignment plans.

Recent studies have focused on the mechanisms, predictive models, and mitigation measures of road traffic noise. Ahmed Elkafoury et al. ^[4] employed the response surface method to develop a traffic noise prediction model, revealing that Leq increases more rapidly when traffic flow is below 1204 veh/h, with notable peaks resulting from interactions between flow-speed and heavy vehicle flow. The model ($R^2=95.87\%$) suggests maintaining noise levels below 65 dB(A) for effective decision-making. Abhijit Debnath et al. ^[5] conducted a study on traffic noise in Dhanbad using 15 monitoring stations. They developed an artificial neural network (ANN) model with a 7-7-5 architecture, utilizing traffic parameters as inputs and LAeq as the output. The method demonstrates high accuracy (± 0.6 dB(A), $R^2>0.9$), surpassing alternative approaches, with regional noise levels exceeding limits and peaking at speeds of 50-55 km/h. Bhagwat Singh Chauhan et al. ^[6] developed prediction models for LAeq and L10 utilizing regression, neural networks, and machine learning, based on data from over 200 sites in Delhi-NCR. Machine learning models demonstrate significant effectiveness (± 3 dB(A)) in noise abatement, providing the first ISO-compliant estimation of hourly noise uncertainty. Domestic research emphasizes the

innovation of prediction models. Chen et al. [7] developed a traffic flow noise prediction model specifically designed for mountainous urban areas by systematically analyzing the acoustic characteristics of hilly cities. Huang et al. [8] introduced a hybrid framework for predicting traffic noise that combines Bayesian Networks (BN) with Partial Least Squares Structural Equation Modeling (PLS-SEM). Shi et al. [9] developed a noise prediction model specific to freeways by incorporating the concepts of equivalent speed and traffic volume, and by translating parameters of large and medium vehicles into metrics equivalent to small vehicles. Qiao et al. [10] conducted a quantitative analysis of the noise reduction efficacy of sound barriers during highway operation phases, utilizing empirical data to propose targeted optimization strategies. Cui et al. [11] developed a composite noise barrier system that integrates sound-absorbing panels, multi-cone absorbers, transparent sound-insulating structures, and quadratic residue diffusers. Zhang et al. [12] utilized the comprehensive pollution index method to assess environmental noise in the urban core of Quanzhou from 2017 to 2022, systematically evaluating regional ambient noise, traffic noise, and zonal functional noise.

In the environmental assessment practice of domestic road engineering, prevalent noise prediction models are based on two principal standards: the traffic noise model specified in the Specification for Environmental Impact Assessment of Highway Construction Projects (JTGB03-2006) by the Ministry of Transport^[13], and the noise prediction framework endorsed in the Technical Guidelines for Environmental Impact Assessment of Highway Construction Projects (HJ 1358-2024) by the Ministry of Ecology and Environment^[14]. These models have historically functioned as essential instruments, although their shortcomings have become increasingly apparent in light of changing planning requirements.

Current traffic noise prediction methods do not account for the noise tolerance thresholds of various acoustic-sensitive sensors. There are significant variations in noise-bearing capacities across different receptor types (e.g., hospitals, schools, residential zones), making the traditional equivalent sound level metric unsuitable for defining real impact severity. Furthermore, current techniques overlook the building scale of affected sensitive receptors and the exposed people density along vehicle corridors. Given that the impacts of noise pollution are determined not only by decibel levels but also by regional propagation ranges and demographic exposure, this oversight adds systematic bias in assessment results. Most importantly, existing frameworks lack the mathematical capacity to evaluate cumulative noise impacts throughout whole routes, nor do they allow for scientifically rigorous comparisons of environmental repercussions between various alignments. These restrictions become most apparent during actual road planning and alignment optimization decision-making.

Building on current road traffic noise forecast models, evaluation methodologies, and proprietary technologies, this study made novel advances by creating a road planning noise assessment model based on the Route Noise Impact Index. The model allows for more detailed noise effect analysis for acoustically sensitive receptors along planned routes by systematically integrating multidimensional data such as regional noise tolerance thresholds, affected building areas for various sensitive receptors, and exposed population size. This approach allows for a quantitative comparison of acoustic environmental impacts across numerous route alternatives, improving the scientific rigor and accuracy of traffic noise assessment methodologies and offering more dependable technical assistance for route alignment optimization.

2. ROAD TRAFFIC OPERATION NOISE PREDICTION MODEL

This study utilizes the noise prediction model specified in the Environmental Impact Assessment of Highway Construction Projects (JTGB03-2006) to assess road traffic operational noise^[13]. This model is predominantly designed for highways with six or fewer lanes (in both directions), first-class highways, and second-class highways, however its structure can also be used to other road classifications.

$$L_{Aeq}(h)_i = (\overline{L_{0E}})_i + 10 \lg\left(\frac{N_i}{V_i T}\right) + 10 \lg\left(\frac{7.5}{r}\right) + 10 \lg\left(\frac{\theta}{\pi}\right) + \Delta L_{16} \quad (1)$$

where: $L_{Aeq}(h)_i$ denotes the hourly equivalent sound level of class i cars (in dB(A)), a crucial parameter that consolidates noise intensity over time.

$(\overline{L_{0E}})_i$ denotes the average radiated noise level of class i vehicles at a horizontal distance of 7.5 meters (in dB(A)); for this study, calculations of such levels for large, medium, and small vehicles follow the formulaic method detailed in Appendix B of the Specification, with vehicle classification criteria specified in Table 1.

V_i stands for the average speed of class i vehicles (in km/h), a variable directly linked to noise generation—faster speeds often correlate with higher emissions. Predictions of this average speed adhere to the computational approach in Appendix C of the Specification, ensuring alignment with road design and traffic flow characteristics.

N_i refers to the average hourly traffic volume of class i vehicles passing the prediction point (in vehicles/hour), distinguishing between daytime and nighttime. This value is derived by converting the proposed highway's segment-specific, year-characteristic traffic volumes using projected vehicle type ratios and the standard day-night ratio (2.5:1), reflecting the stark difference in noise impact between peak and off-peak hours.

R measures the distance from the lane centerline to the prediction point (in meters), with the formula applicable only when $R > 7.5$ meters—a constraint rooted in simplifications of near-field noise propagation physics. T is the duration for calculating the equivalent sound level, set here to 1 hour, consistent with standard noise assessment protocols.

θ represents the angle subtended by the finite-length road segment at the prediction point (in radians), a parameter that adjusts for how road length influences noise spread; its determination follows Figure 1 in Appendix B of the Specification.

ΔL accounts for corrections from other factors (in dB(A)), such as terrain or barriers, refining predictions to match real-world conditions.

Table 1. Classification of road traffic noise vehicle types.

<i>vehicle type</i>	<i>representative model</i>	<i>Vehicle Classification Criteria</i>
small	minibus	Buses ≤ 19 seats and trucks $\leq 2t$ capacity
medium	omnibus	Buses with >19 seats and trucks with $>2t$
Large	large vehicle automobile train	$7t < \text{Load capacity} \leq 20t$ trucks, Trucks with a capacity of $>20t$

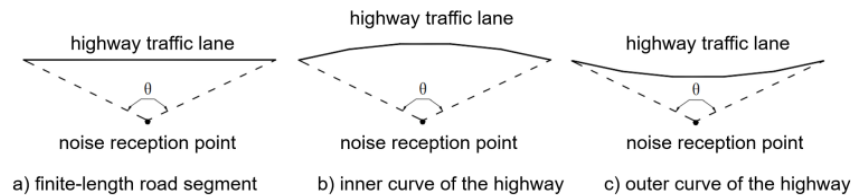


Figure 1. The angle subtended by the two ends of a finite-length road segment at the prediction point.

Total Traffic Equivalent Sound Level:

$$L_{Aeqg}(T) = 10 \lg(10^{0.1L_{Aeq}(h)^L} + 10^{0.1L_{Aeq}(h)^M} + 10^{0.1L_{Aeq}(h)^S}) \quad (2)$$

where: $L_{Aeqg}(T)$ is the noise contribution value generated by the sound source of the highway project at the prediction point, dB(A).

$L_{Aeq}(h)^L$, $L_{Aeq}(h)^M$, $L_{Aeq}(h)^S$ are noise contribution values, dB(A), for large, medium and small vehicles respectively.

If a prediction point is affected by noise from several roads, the sound level of each route should be calculated separately for the prediction point and then superimposed to obtain the noise contribution value. After obtaining the noise contribution value, it is necessary to correct the noise contribution value, such as the longitudinal slope of the road, the reflection of the buildings on both sides, etc. will affect the noise contribution value.

Based on the noise contribution value and background noise, determine the predicted noise value of the route to be planned at the sensitive points of the acoustic environment, and use the following formula to calculate the predicted noise value:

$$L_{NP} = 10 \lg(10^{0.1L_{NCV}} + 10^{0.1L_{BN}}) \quad (3)$$

where: L_{NP} is noise prediction value.

L_{NCV} is the total traffic equivalent sound level is used as the noise contribution value.

L_{BN} is use of status quo monitoring values or analogous status quo monitoring values at the same point as the current state of the environment as background values.

2.1. Evaluation methodology

The systematic classification of acoustic functional zoning constitutes a critical prerequisite for rigorous noise impact evaluation. Building upon the acoustic environmental functional zoning framework specified in the Technical Specifications for Acoustic Environmental Functional Zoning (GB/T 15190-2014)^[15], this study established a building classification system for traffic planning noise impact assessment through systematic integration of domain-specific expertise (including 5 senior experts in environmental acoustics and urban-rural planning) and empirical project investigation data. As presented in Table 2, the established hierarchical noise sensitivity classification matrix categorizes built environments into five distinct classes with differentiated permissible noise ceilings, incorporating temporal variations between daytime (06:00-22:00) and nighttime (22:00-06:00) operational periods.

Table 2. Sensitive points corresponding to the acoustic environment building category corresponding to the Maximum permissible standard noise.

<i>Acoustic environment building type</i>	<i>daytime</i>	<i>Nighttime</i>
Category 0	50dB(A)	40dB(A)
Category 1	55dB(A)	45dB(A)
Category 2	60dB(A)	50dB(A)
Category 3	65dB(A)	55dB(A)
Category 4	70dB(A)	55dB(A)

Class 0 Acoustic Environmental Buildings: Designate architectural units requiring special quiet protection, such as rehabilitation sanatoriums and high-end quiet-care medical facilities, necessitating stringent control of background noise interference.

Class 1 Acoustic Environmental Buildings: Encompass residential communities, research institutions, non-rehabilitation medical facilities, educational institutions, and administrative offices, where daily living, academic, or administrative functions mandate maintained ambient quietness.

Class 2 Acoustic Environmental Buildings: Include commercial complexes, financial districts, marketplaces, and mixed commercial-industrial zones, permitting moderate noise levels within defined thresholds while ensuring acoustic manageability.

Class 3 Acoustic Environmental Buildings: Primarily industrial production zones and logistics parks, characterized by elevated noise tolerance yet requiring compliance with occupational health and safety standards.

Class 4 Acoustic Environmental Buildings: mainly refers to other buildings that need to avoid the serious impact of road traffic noise on the surrounding environment.

This classification system ensures the scientific validity and applicability of noise standards through dynamic adjustments that integrate on-site surveys of building usage, population density, and characteristics of surrounding sound sources, combined with regulatory requirements.

Quantitative assessment of noise pollution must integrate both exceedance magnitude and socio-environmental impact dimensions. This study defines noise exceedance as the dynamic differential between predicted noise levels at acoustic-sensitive receptors (ASRs) and the corresponding standard limits for acoustic environmental building categories, quantified through equivalent continuous A-weighted sound pressure level ($L_{eq, A}$) measurements:

$$\Delta L_{Aeq}(j) = L_{NP}(j) - L_s(j) \quad (4)$$

where: $\Delta L_{Aeq}(j)$ is noise exceedance of the j th building in dB.

$L_{NP}(j)$ is predicted noise level of the j -th building, in dB(A), obtained via the Cadna/A noise prediction model or field measurements.

$L_s(j)$ is standard noise limit for the acoustic environmental building category corresponding to the j -th building in Table 2, in dB(A).

The noise exceedance level transcends conventional single-threshold evaluations, explicitly characterizing pollution status through positive/negative values (positive indicating exceedance, negative denoting compliance), thereby providing foundational data for subsequent comprehensive assessments.

Building upon the exceedance metrics, this study developed the Route Noise Impact Index (RNII), which innovatively couples noise exposure intensity with affected population size and spatial dimensions to establish a multidimensional evaluation system^[16]. The methodology involves three operational phases: (1) determining building-specific noise impact indices for each acoustically sensitive receptor along proposed routes, (2) integrating corridor-wide impacts through spatial weighting algorithms, and (3) generating comprehensive RNII values quantifying total route noise impacts. The computational framework is expressed as:

$$RNCII = \sum_{j=1}^n \Delta L_{Aeq}(j) \times P_j \times S_j \quad (5)$$

where: RNCII is route noise impact index in dB·person·m².

P_j is the number of people working, studying, or living in the j th building affected by noise, in persons.

S_j is floor area of the j th building affected by noise, in m².

n is the total number of buildings in the traffic noise impact area along the proposed project.

This evaluation framework transcends the limitations of conventional noise assessments that focus solely on physical parameter exceedances. By introducing socio-spatial parameters such as population density and building floor area ratio, we established a tripartite association model integrating "noise exposure-spatial carriers-population response", thereby providing a more practical quantitative tool for evaluating the environmental benefits of transportation planning schemes.

3. CASE APPLICATIONS

A first-level highway in Guangxi Province, with a design speed of 80 km/h and a standard roadway width of 26.0 meters, features an asphalt concrete pavement. When passing through a certain township, two route alignment options were proposed. Route alignment option 1 passes south of the township, while route alignment 2 passes north of the township. Both route alignment options have similar and relatively gentle longitudinal slopes. Route alignment 1 measures approximately 4.7 km in length, and route alignment 2 measures approximately 4.5 km. Based on the engineering quantities and investment estimates, the construction cost of route alignment 1 is approximately RMB 650,000 more than that of route alignment 2. However, route alignment 2 is closer to more noise-sensitive points, with denser buildings and a higher population along its corridor. By applying the traffic noise impact assessment method proposed in this study, a more scientific and comprehensive analysis of the advantages, disadvantages, and comprehensive costs of the two route alignment options was conducted.

Table 3. Calculation of Noise Impact Index for Option 1-Route.

<i>Sensitive site number</i>	<i>Acoustic environment building type</i>	<i>Building area (m²)</i>	<i>population</i>	<i>Noise background value (daytime/dB)</i>	<i>Noise background value (night/dB)</i>	<i>Noise prediction value (daytime/dB)</i>	<i>Noise prediction (night/dB)</i>	<i>Noise impact index (daytime/dB·person·m)</i>	<i>Noise impact index (night/dB·person·m)</i>	<i>Noise prevention and control measures</i>
1	1	275	6	53.2	44.3	59.4	50.3	7260	8745	<i>Measure 1+Measure 2</i>
2	1	210	5	46.3	39.0	58.1	49.4	3255	4620	<i>Measure 1+Measure 2</i>
3	3	365	2	54.7	42.3	60.5	50.6	0	0	<i>Measure 2</i>
...	...	...	...	...	...	...	...	...	...	...
15	2	140	6	53.0	43.5	59.9	49.8	0	0	<i>Measure 2</i>
16	1	210	4	52.8	44.5	57.4	48.1	2016	2604	<i>Measure 2</i>
Cumulative Route Noise Impact Index (dB·person·m²)								37460	46310	

Table 4. Calculation of Noise Impact Index for Option 2-Route.

<i>Sensitive site number</i>	<i>Acoustic environment building type</i>	<i>Building area (m²)</i>	<i>population</i>	<i>Noise background value (daytime/dB)</i>	<i>Noise background value (night/dB)</i>	<i>Noise prediction value (daytime/dB)</i>	<i>Noise prediction (night/dB)</i>	<i>Noise impact index (daytime/dB ·person·m)</i>	<i>Noise impact index (night/dB ·person·m)</i>	<i>Noise prevention and control measures</i>
1	1	324	5	52.8	45.6	59.9	51.2	7938	10044	<i>Measure 1+Measure 2</i>
2	2	500	8	50.1	42.6	59.4	50.3	0	1200	<i>Measure 1+Measure 2</i>
3	3	385	6	53.2	44.2	61.2	51.8	14322	15708	<i>Measure 1+Measure 2</i>
...	...	...	...	...	...	...	...	...	...	...
26	2	260	4	52.8	42.9	60.3	50.4	0	0	<i>Measure 1+Measure 2</i>
27	1	390	7	54.3	46.7	59.1	50.2	11193	14196	<i>Measure 1+Measure 2</i>
Cumulative Route Noise Impact Index (dB ·person·m²)								70875	102137	

Note: (1).The predicted road traffic noise values in the table can be analyzed by using German Cadna/A (the software has been certified by the Environmental Engineering Evaluation Center of the Ministry of Environmental Protection and is considered suitable for the simulation and prediction of the acoustic environment in China) or other environmental noise simulation software for noise prediction.(2).Measure 1—Speed limit + horn ban, Measure 2—Install soundproof windows.

According to the survey and calculation results, as shown in Table 3, route alignment 1 passes through fewer noise-sensitive buildings, with a total affected building area of approximately 3,440 square meters and an affected population of about 80 people. It has 16 noise-sensitive points exceeding the limit, with noise levels ranging from 47.8 dB(A) to 70.9 dB(A). The daytime traffic noise impact index is 37,460 dB, and the nighttime traffic noise impact index is 46,310 dB. The traffic noise impact of route alignment 1 is relatively small. To ensure that the predicted noise levels of the noise-sensitive buildings along the alignment remain at their original levels or meet indoor noise functional requirements, the following primary control measures are recommended: (1) installation of speed limit and silent zone signs at critical noise-sensitive sections and nearby road segments; and (2) replacement of soundproof windows for a small number of buildings adjacent to the highway. The comprehensive noise control cost for route alignment 1 is approximately RMB 320,000.

In comparison, as shown in Table 4, route alignment 2 passes through more noise-sensitive buildings, with a total affected building area of approximately 6,950 square meters and an affected population of about 121 people. It has 27 noise-sensitive points exceeding the limit, with noise levels ranging from 48.8 dB(A) to 73.2 dB(A). The daytime traffic noise impact index is 70,875 dB (1.89 times that of route alignment 1), and the nighttime traffic noise impact index is 102,137 dB (2.20 times that of route alignment 1). The traffic noise impact of route alignment 2 is significantly greater. To ensure that the predicted noise levels of the noise-sensitive buildings along the alignment remain at their original levels or meet indoor noise functional requirements, the following measures are required: (1) installation of speed limit and silent zone signs at critical noise-sensitive sections and nearby road segments; (2) installation of white noise barriers at critical sensitive sections; and (3) replacement of soundproof windows for a larger number of buildings adjacent to the highway.

Quantitative comparisons between the two route alternatives reveal significant differences in technical and economic parameters. As indicated in Tables 5 and 6, Route Option 1 demonstrates a construction and installation cost exceeding that of Route Option 2 by approximately 650,000 CNY. However, the comprehensive noise control expenditure for Option 2 surpasses Option 1 by 970,000 CNY. When integrating noise mitigation measures into total project costs, Route Option 2 exhibits marginally higher (3.2%) overall construction expenditure compared to Option 1.

Table 5. Indicators for programmatic evaluations.

<i>Evaluation Indicators</i>	<i>Route Program 1</i>	<i>Route Program 2</i>
Population affected	80 people	185 people
Impact on floor area	3340(m ²)	6950(m ²)
Comprehensive construction cost	18500 (10,000RMB)	18532 (10,000 RMB)
Route noise impact index (daytime)	37460 (dB ·person·m)	70875 (dB ·person·m)
Route noise impact index (night)	46310 (dB ·person·m ²)	102137 (dB ·person·m ²)

This normalization process resolves the inherent incomparability between metrics of different units. For instance, "Comprehensive Construction Cost" (monetary units) and "Affected Population" (cardinal numbers) become directly comparable post-normalization. The transformed data reveals clearer relative performance patterns. Taking "Route Noise Impact Index (Daytime)" as an example, the pre-normalization values of 42.8 dB(A) (Option 1) and 58.3 dB(A) (Option 2) translate to normalized scores of 0.35 and 0.65 respectively, providing immediate visual differentiation of technical advantages. As shown in Figure 2, the radar chart of programme evaluation indicator values intuitively presents the overall distribution of these normalized scores across both route options.

Table 6. Evaluation indicators normalized to sum total.

<i>Evaluation Indicators</i>	<i>Route Program 1</i>	<i>Route Program 2</i>	<i>Direction of Indicators</i>
Population affected	0.30	0.70	The smaller the better
Impact on floor area	0.32	0.68	
Comprehensive construction cost	0.50	0.50	
Route noise impact index (daytime)	0.35	0.65	
Route noise impact index (night)	0.31	0.69	

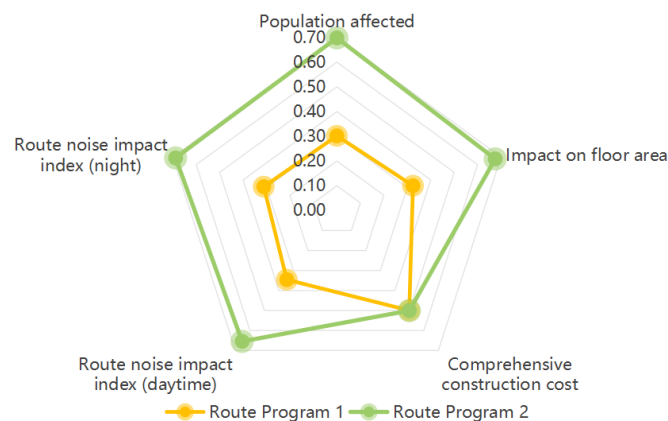


Figure 2. Radar chart of programme evaluation indicator values.

The systematic evaluation demonstrates that Route Option 1 achieves better balance between economic efficiency and environmental impact mitigation. Based on these findings, it is recommended to adopt Route Option 1 as the preferred implementation plan for this project.

4. CONCLUSIONS

China presently lacks a dedicated regulatory framework especially addressing vehicle traffic noise emissions. Current traffic noise prediction analyses often utilize single-point equivalent continuous A-weighted sound level (L_{eq}) evaluations to ascertain noise level exceedances. This traditional single-point assessment model demonstrates many constraints: Firstly, its singular analysis of noise exposure at particular receptors overlooks the spatial continuous attributes of noise effects along linear engineering projects. Secondly, the insufficient evaluation of essential factors, including the magnitude of impacted people and variances in functional categories among sensitive locations, leads to inappropriate integration of noise impact assessments with urban planning and transportation design procedures.

This study's Road Noise Impact Index (RNII) methodology overcomes these shortcomings via two primary innovations:

① Establish tiered threshold criteria based on noise sensitivity classes, and ② Integrate spatial information such as population density and building footprints to convert single-point exceedance assessments into full corridor-level effect analysis. This improved method allows for accurate detection of "concealed non-compliance areas" resulting from receptor clustering effects, offering quantitative decision-making assistance for optimizing new route alignments and retrofitting existing road noise mitigation measures.

The measurement of road traffic noise during the operational phase presents a complicated technical task, involving interrelated factors such as traffic flow dynamics, sound propagation circumstances, and receptor response characteristics. Future research should focus on: ① Enhancing highway traffic noise prediction models, and ② Establishing context-specific noise thresholds for various auditory environment categories. These developments will enhance the theoretical basis for noise-informed route selection frameworks and construct a cohesive "prediction-assessment-optimization-management" technical system. This accomplishment will significantly facilitate the advancement of transportation infrastructure towards more sustainable growth and improved human-centricity.

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